

Practical # 2: Maven + Docker Integration

Step 1: Create Maven Project

```
mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo  
-DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

Step 2: Package Application

Mvn clean package

Step 3: Modify Java Program

With path : src/main/java/com/example/App.java

Step 4: Create Dockerfile

```
FROM eclipse-temurin:21-jdk  
  
WORKDIR /app  
  
COPY target/docker-maven-demo-1.0-SNAPSHOT.jar app.jar  
  
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Step 4: Add dockerfile-maven-plugin in pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
    http://maven.apache.org/xsd/maven-4.0.0.xsd">  
  
  <modelVersion>4.0.0</modelVersion>
```

<groupId>com.example</groupId>

<artifactId>docker-maven-demo</artifactId>

<version>1.0-SNAPSHOT</version>

<packaging>jar</packaging>

<name>docker-maven-demo</name>

<url>http://maven.apache.org</url>

<properties>

 <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>

 <maven.compiler.source>21</maven.compiler.source>

 <maven.compiler.target>21</maven.compiler.target>

</properties>

<dependencies>

 <!-- JUnit for testing -->

 <dependency>

 <groupId>junit</groupId>

 <artifactId>junit</artifactId>

 <version>4.13.2</version>

 <scope>test</scope>

 </dependency>

</dependencies>

<build>

 <plugins>

 <plugin>

 <groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-jar-plugin</artifactId>

<version>3.3.0</version>

<configuration>

<archive>

<manifest>

<mainClass>com.example.App</mainClass>

</manifest>

</archive>

</configuration>

</plugin>

<!-- Dockerfile Maven Plugin -->

<plugin>

<groupId>com.spotify</groupId>

<artifactId>dockerfile-maven-plugin</artifactId>

<version>1.4.13</version>

<configuration>

<repository>docker-maven-demo</repository>

<tag>\${project.version}</tag>

</configuration>

<executions>

<execution>

<id>default</id>

<goals>

<goal>build</goal>

</goals>

</execution>

</executions>

</plugin>

```
</plugins>
```

```
</build>
```

```
</project>
```

Step 5: Build Docker Image Using Maven

```
mvn dockerfile:build
```

Step 6: Verify Docker Image

Docker images

Step 7: Run Docker Container

```
docker run docker-maven-demo:1.0-SNAPSHOT
```

